HiLoc: A Hierarchical Representation Method for Spatial Localization in Multimodal Large Language Models

Evelyn Zhang[1,2,†] **Fufu Yu**[1] **Hanjun Li**[1] **Aoqi Wu**[3]
Ke Yan[1,*] **Shouhong Ding**[1] **Tianhe Ren**[4] **Xinting Hu**[4]
Xiaojuan Qi[4,*] **Li Jiang**[5] **Jiaya Jia**[2]

[1]Tencent Youtu Lab
[2]The Hong Kong University of Science and Technology
[3]The Hong Kong Polytechnic University [4]The University of Hong Kong
[5]The Chinese University of Hong Kong, Shenzhen

Abstract

Existing multimodal large language models (MLLMs) typically formulate object detection as a conditional sequence generation task. Under this paradigm, spatial locations are commonly represented by either textual coordinates or special quantized learnable tokens. However, textual coordinates are tokenized as unrelated symbols, spatially adjacent points (e.g., $x = 199$ vs. $y = 200$) yield entirely different tokens, so the cross-entropy loss is misaligned with true spatial distance. Special quantized location tokens, on the other hand, inevitably introduce quantization error. Reducing this error requires linearly more tokens, but a larger vocabulary is harder to train and converges slower. We argue that both limitations stem from the formulation of spatial representation. To address this issue, we propose a simple yet novel localization representation method, **HiLoc**, which predicts locations in a coarse-to-fine manner with special tokens: first a grid index for coarse localization, then a cell index to refine the position within the grid. Under this paradigm, accuracy can be scaled up by simply adding the hierarchy level. We further rigorously prove that, with fewer learnable special tokens, HiLoc yields smaller quantization error, more stable training, and limited error propagation. We also build **HiRef-500K**, a dataset consisting negative and complex referring samples to strengthen rejection and complex referring ability. Experiments show that our method achieves strong performance on both detection and grounding benchmarks, especially on small objects, under high IOU thresholds and complex referring tasks.

Project page: `https://yuweizhang2002.github.io/HiLoc/`.

1 Introduction

Object detection is a fundamental capability in computer vision, underpinning a wide range of real-world applications. Traditional detectors formulate localization as a *coordinate regression* problem, where dense anchors or learnable queries are decoded into bounding boxes via regression heads trained with IoU-based losses [2, 7, 21, 25, 58]. But they rely on specialized architectures, hindering unification with free-form language understanding and reasoning.

*Co-corresponding author
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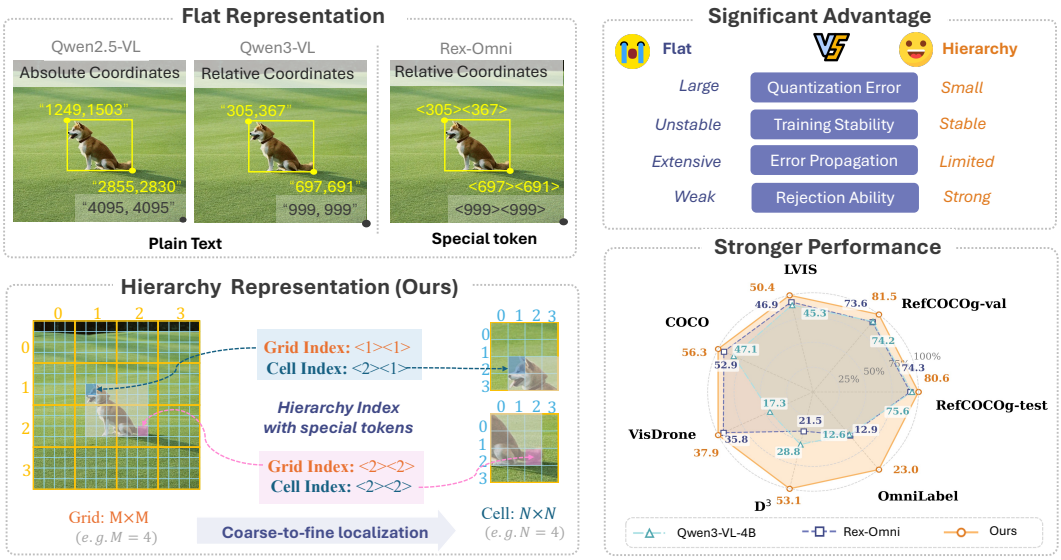


Figure 1: **Overview of HiLoc.** Existing MLLMs encode a location as a flat sequence of coordinate tokens (plain-text [1, 6, 47] or special tokens [16]). HiLoc instead predicts a coarse-to-fine hierarchical spatial index with special tokens, yielding small quantization error, more stable training, limited error propagation, and strong rejection ability for absent targets. HiLoc consistently surpassing strong baselines across diverse detection and grounding benchmarks.

The rapid progress of MLLMs [1, 13, 27, 41, 42, 47] offers an appealing alternative: jointly performing understanding and localization by recasting localization as a *language modeling* problem, expressing each location as a flat sequence of coordinate tokens generated autoregressively under token-level cross-entropy supervision during training [4, 5, 16, 31, 50, 53]. To represent the coordinate, a straightforward approach is to use *plain-text digits*. However, plain digits carry no spatial meaning after tokenization: as illustrated in Figure 2(b1), two spatially close points in L1 distance may produce token sequences whose cross-entropy loss is unexpectedly large after tokenization (e.g., "299" and "305"). To inject spatial inductive bias, Rex-Omni [16] introduces a large set of learnable special tokens (e.g., <0> $\sim$ <999>) that index locations. It learns the underlying positional relationships among special tokens through extensive training and large detection datasets. This further leads to a trade-off: *localization accuracy increases linearly with the number of learnable special tokens, yet more tokens are much harder to learn.*

We argue that the root cause is treating 2D space as an *unstructured sequence of tokens*, whereas human spatial reasoning is naturally *coarse-to-fine*—first locating a rough region, then refining within it. Motivated by this observation, we propose **HiLoc**, which reformulates spatial localization as *coarse-to-fine hierarchical index prediction* with a small set of learnable special tokens. Shown in Figure 2(a), HiLoc first divides the image into a coarse $M \times M$ grid, then further partitions each grid into $N \times N$ fine-grained cells. The location is represented by both the grid and cell indices. The two levels share the same small speical token set, *so localization accuracy scales quadratically with the length of learnable special tokens, making it easier to learn latent spatial positional relationships.*

As shown in Figure 2(b), HiLoc's design offers the following four advantages: **(i) Better geometry awareness**: neighboring coordinates are likely to share the same grid prefix, so coordinates that are spatially close will also be closer in the embedding space. **(ii) Smaller error propagation**: a single mis-predicted cell token index causes at most a less than one-grid drift in HiLoc, whereas a wrong leading digit of a flat coordinate induces an approximately 100px jump and a large IoU drop. **(iii) More stable training**: suppose the image has height H and width W, and let $\max(H, W) = L^2$. Under the same resolution, HiLoc needs only L special tokens, while Rex-Omni requires L^2 special tokens. HiLoc's smaller classification space (L vs. L^2) yields more stable cross-entropy optimization. **(iv) Less quantization error**: with the same number of special tokens L, HiLoc's per-axis quantization error is $W/(2L^2)$, considerably smaller than the $W/(2L)$ of Rex-Omni.

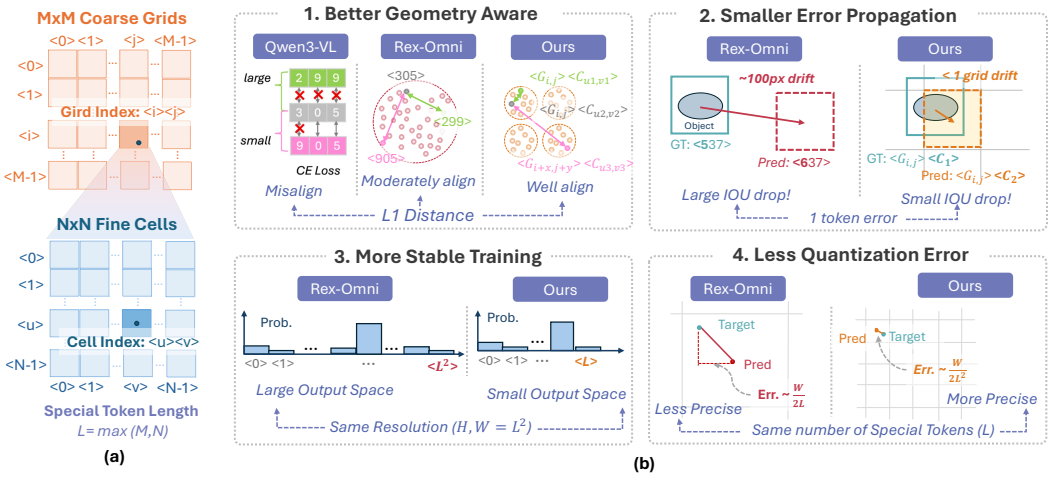


Figure 2: **(a)** Illustration of the coarse-to-fine hierarchical spatial index location representation method proposed by HiLoc. Space is factorized into an $M \times M$ coarse grid $\langle G_{i,j} \rangle$ and an $N \times N$ fine cell $\langle C_{u,v} \rangle$, sharing a single vocabulary of size $L = \max(M, N)$. **(b)** HiLoc's hierarchical indexing brings four main advantages over flat coordinate representation methods [16, 47], including better geometry aware, small error propagation, more stable training and less quntization error.

Moreover, existing MLLMs suffer from poor rejection capability, largely due to the lack of datasets with negative examples and complex referring expressions. We thus propose **HiRef-500K**, a dataset of about 500K annotated expressions covering five types of complex expressions, including attribute, state, relation, set logic, and non-exist.

In summary, our contributions are threefold:

- We propose **HiLoc**, which formulate localization as coarse-to-fine hierarchical index prediction with a small set of special tokens, yielding better geometry awareness, smaller error propagation, more stable training, and less quantization error.
- We construct **HiRef-500K**, a referring expression dataset with diverse phrases and negative examples, which improves complex understanding and rejection capability of the model.
- Experiments show our method excels on detection and grounding benchmarks, particularly under high IoU, small objects, and complex referring tasks.

2 Related Work

Regression-based and Open-Vocabulary Detection. Object detection has evolved from anchor-based CNN models such as Faster R-CNN [12, 33] and SSD [26], through anchor-free designs like FCOS [39], to Transformer-based detectors including DETR [2] and Deformable DETR [58], with continuous improvements in head design and optimization [8, 24]. To move beyond fixed category sets, open-vocabulary methods leverage vision-language alignment to enable zero-shot recognition. Representative works include GLIP [21], GroundingDINO [25], YOLO-World [7], and DINO-X [34], which progressively broaden the recognizable concept space through text-prompted or visual-prompted paradigms. These detector-style approaches optimize IoU-based regression losses, yielding high spatial precision. However, they rely on task-specific architectures and lack the deep language understanding needed for context-rich, real-world scenarios such as complex referring expressions and open-ended descriptions.

MLLM-based Localization and Coordinate Representation. In contrast, MLLM-based methods leverage the strong semantic reasoning of large language models to tackle complex localization, differing in how spatial outputs are produced. Existing approaches can be grouped into three paradigms. (1) *Retrieval-based methods* [28, 49] incorporate an external proposal module and train the LLM to select among candidate regions, with ROD-MLLM [49] adding rejection-aware retrieval to suppress false positives; however, these methods depend on the quality and coverage of the proposal set, limiting their flexibility in open-world settings. (2) *External-decoder methods* [20] map

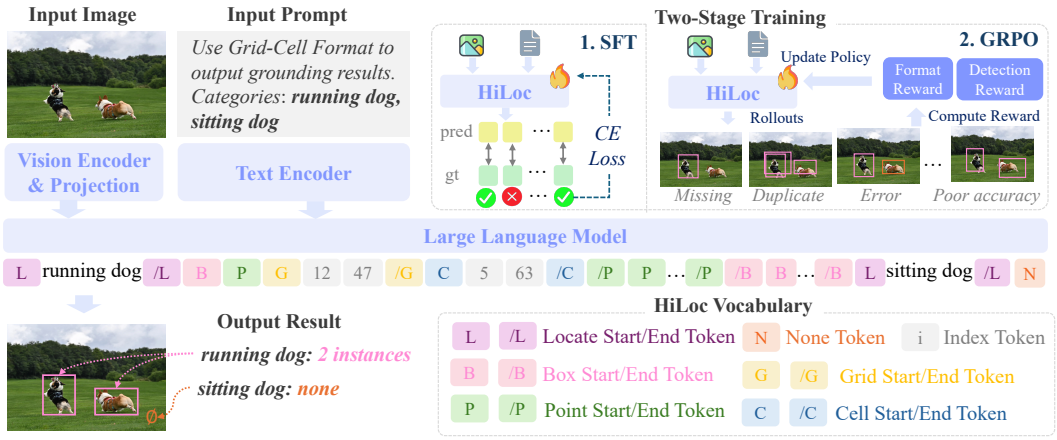


Figure 3: **Training Pipeline of HiLoc.** Each bounding box is encoded as two points (top-left and bottom-right), each point is decomposed into a coarse `Grid` index and a fine `Cell` index. Additionally, a none token is introduced to explicitly support rejection. The model is constructed based on Qwen3VL-4B and trained with SFT, followed by GRPO post-training.

special LLM embeddings to an auxiliary head for final coordinate output, but introduce additional architectural complexity and require extra supervision signals. (3) *Direct coordinate prediction* [4, 5, 16, 31, 41, 50, 53, 54] treats coordinate values as discrete tokens within the LLM vocabulary and generates them autoregressively, offering a simple, end-to-end formulation without relying on external modules or additional supervision, and has been widely adopted by general-purpose MLLMs [1, 13, 27, 42, 47], with Rex-Omni [16] further introducing special coordinate tokens and reinforcement learning. Within this dominant paradigm, a key design choice is *how coordinates are tokenized*. There are three variants: (i) *absolute coordinates without special tokens* [1], which is expressed directly using absolute numerical values in plain text, incurring long sequences after tokenization, especially when the numerical values are large; (ii) *relative coordinates without special tokens* [13], where the image dimensions are normalized to a fixed range (e.g., "0" $\sim$ "999"), and coordinates are expressed as plain-text numerical values within that normalized space; and (iii) *relative coordinates with special tokens* [5, 16], where the normalized relative coordinates are represented using special discrete learnable tokens (e.g., `<0>` $\sim$ `<999>`), yielding the compact tokenized sequence at the cost of expanding the learnable vocabulary.

3 Method

3.1 Overview of HiLoc

As illustrated in Figure 3, HiLoc reformulates localization task as *hierarchical spatial index prediction*. We instantiate HiLoc on Qwen3-VL-4B-Instruct [47] by extending its tokenizer with a small set of discrete index tokens and structured tags. It supports multi-phrase, multi-instance localization with structured phrase grouping, and integrates a dedicated `<no_box>` rejection token for phrases with no valid instance. Training proceeds in two stages: supervised fine-tuning (SFT) to learn the output format and basic localization behavior, followed by GRPO post-training with detection-quality and format-compliance rewards to further improve accuracy and reliability.

Hierarchical Spatial Indexing. Given an image of size $W \times H$, we impose an $M \times M$ coarse grid (grid size G_w, G_h), each further divided into $N \times N$ fine cells (cell size C_w, C_h). A point $P=(P_w, P_h)$ is encoded by its grid index (G_i, G_j) and cell index (C_u, C_v):

$$G_i = \left\lfloor \frac{P_w}{G_w} \right\rfloor, \; G_j = \left\lfloor \frac{P_h}{G_h} \right\rfloor, \quad C_u = \left\lfloor \frac{P_w - G_i G_w}{C_w} \right\rfloor, \; C_v = \left\lfloor \frac{P_h - G_j G_h}{C_h} \right\rfloor. \tag{1}$$

This decomposes localization into coarse region selection followed by local refinement. All indices are represented as discrete special tokens $\langle 0 \rangle, \ldots, \langle K-1 \rangle$ (shared across Grid and Cell; $K = \max(M, N)$), replacing textual coordinates and yielding a compact, stable CE objective over bounded categories.

Box Representation and Structured Output HiLoc adopts a markup-style output grammar for reliable parsing. Each query phrase is wrapped by ... , followed by zero or more <box>...</box> predictions. Each box consists of two hierarchical points (top-left P^{tl} and bottom-right P^{br}), where a <point> nests <grid>(G_i, G_j) and <cell>(C_u, C_v) fields using index tokens:

$$\texttt{<point> <grid>}\ \langle G_i\rangle\langle G_j\rangle\ \texttt{</grid> <cell>}\ \langle C_u\rangle\langle C_v\rangle\ \texttt{</cell> </point>}. \tag{2}$$

If no instance exists, a dedicated <no_box> token is emitted. For multi-phrase prompts, phrase groups are concatenated sequentially, each followed by its boxes or <no_box>.

3.2 Theoretical Analysis

In this section, we formally analyze HiLoc's advantages over flat coordinate generation. Consider an image of size $W\times H$ with an $M\times M$ coarse grid and $N\times N$ fine cells per grid. HiLoc achieves equivalent precision with exponentially fewer special token, more stable trainning, and limited error propagation. Full proofs are in Appendix A. Additional assumptions and corrections to these statements are given in Appendix H.

Proposition 3.1 (Quantization Error)**.** *Under a fixed vocabulary budget L, flat methods incur the same per-axis quantization error $W/(2L)$, while HiLoc achieves a quantization error of $W/(2L^2)$ and and thus provides up to $L\times$ finer precision.*

Proposition 3.2 (Training Stability)**.** *Under the same per-axis resolution of $K{=}MN$, the initial CE loss per step is $\log(\max(M,N))$ for HiLoc versus $\log(MN)$ for flat methods, yielding a better-conditioned optimization landscape.*

Proposition 3.3 (Error Propagation)**.** *Let $\hat{P}_w$ denote the predicted x-coordinate, and p_g, p_c the per-axis error probabilities of the Grid and Cell predictions, respectively. The expected per-axis error satisfies:*

$$\mathbb{E}\big[|\hat{P}_w - P_w|\big]_{\text{Hi}} \ \le\ p_g\cdot\frac{W}{M} + (1{-}p_g)\,p_c\cdot\frac{W}{MN} + (1{-}p_g)(1{-}p_c)\cdot\frac{W}{2MN}. \tag{3}$$

The worst-case error (Grid mistake) is $W/M{\approx}0.016W$ for $M{=}64$, versus $0.1W$ for a leading-digit error in flat 3-digit generation, and errors degrade gracefully across levels.

3.3 Training Recipe

Stage 1: SFT. We first train the model with next-token cross-entropy on the structured output format: $\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{t=1}^{T}\log p_\theta(y_t \mid y_{<t}, x)$, where x is the input and y is the target sequence.

Stage 2: Reinforcement Learning with GRPO. We further apply GRPO to improve localization quality. For each input x, we sample G candidate outputs $\{y^{(g)}\}$ from the current policy and score each with a composite reward combining a detection term and a format term: $r(x,y) = \lambda_{\text{det}} r_{\text{det}}(x,y) + \lambda_{\text{fmt}} r_{\text{fmt}}(y)$, where $\lambda_{\text{det}}, \lambda_{\text{fmt}}{\ge}0$ are scalar weights.

(i) Detection reward. For each phrase, let $\mathcal{B}(y)$ and $\mathcal{B}^\star$ denote the predicted and ground-truth box sets. When $|\mathcal{B}^\star|{>}0$, we perform greedy IoU matching with threshold τ, which yields the true/false positives and false negatives $\mathrm{TP}, \mathrm{FP}, \mathrm{FN}$, and compute:

$$r_{\text{det}}^{\text{phrase}} = F_\beta\ -\ \lambda_{\text{fn}}\cdot\frac{\mathrm{FN}}{|\mathcal{B}^\star|}\ -\ \gamma_{\text{fp}}\cdot\frac{\mathrm{FP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\epsilon}, \tag{4}$$

where $F_\beta{=}(1{+}\beta^2)\,\mathrm{TP}\big/\big((1{+}\beta^2)\mathrm{TP}+\beta^2\mathrm{FN}+\mathrm{FP}\big)$ is the standard F-measure with β, that controls the trade-off between precision and recall. $\lambda_{\text{fn}}, \gamma_{\text{fp}}$ penalize missed and spurious detections, and ϵ is a small constant for numerical stability. When $|\mathcal{B}^\star|{=}0$, correct rejection receives a reward r_0, while each false prediction incurs a penalty α:

$$r_{\text{det}}^{\text{phrase}} = \begin{cases} r_0, & \text{if } |\mathcal{B}(y)| = 0,\\ \mathrm{clip}(-\alpha|\mathcal{B}(y)|,\ \eta,\ 0), & \text{otherwise.}\end{cases} \tag{5}$$

where η is the lower bound of the clip function. The overall detection reward averages over the phrase set $\mathcal{P}$ of the sample: $r_{\text{det}} = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} r_{\text{det}}^{\text{phrase}}(p)$.

(ii) Format reward. $r_{\text{fmt}}{\in}\{0,1\}$ enforces syntactic compliance, returning 1 if the output conforms to the structured grammar in Eq. (2) and 0 otherwise.

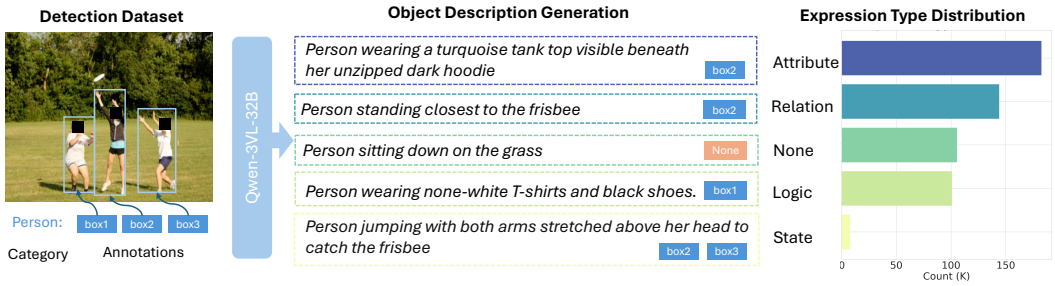


Figure 4: Overview of the data engine for constructing **HiRef-500K**. The image is from LVIS[14].

Table 1: Performance Comparison on COCO Dataset for Common Object Detection

Type	Method	COCO [23]								
		R@IoU$_{0.5}$	P@IoU$_{0.5}$	$F1$@IoU$_{0.5}$	R@IoU$_{0.95}$	P@IoU$_{0.95}$	$F1$@IoU$_{0.95}$	R@mIoU	P@mIoU	$F1$@mIoU
Closed-set	Faster RCNN-R50 [33]	60.4	60.7	60.6	4.9	12.6	7.1	43.2	54.2	48.1
	DETR-R50 [2]	59.6	73.9	65.9	10.6	19.0	13.6	42.9	55.3	48.3
	DyHead-R50 [8]	58.1	76.6	66.1	11.9	20.6	15.0	44.8	60.1	51.3
	DAB-DETR-R50 [24]	59.2	77.4	67.1	10.5	18.5	13.4	43.8	58.8	50.2
	Deformable-DETR-R50 [58]	62.6	78.5	69.7	14.8	17.7	17.7	48.7	62.4	54.7
	DINO-R50 [55]	62.6	76.5	68.8	17.8	25.8	21.1	50.0	62.4	55.6
	DINO-Swin-L [55]	**69.8**	**82.5**	**75.6**	**22.2**	**29.7**	**25.4**	**57.0**	**68.2**	**62.1**
Open-set	Grounding DINO-Swin-T [25]	62.8	78.4	69.8	20.5	26.2	23.0	54.6	58.8	56.6
MLLM	DeepSeek-VL2-Tiny [42]	29.4	55.0	38.2	5.9	13.0	8.1	20.0	38.6	26.3
	OVIS2.5-9B [27]	60.0	45.3	51.6	8.4	8.0	8.2	39.3	31.0	34.6
	Mimo-VL-7B [44]	56.2	56.9	56.5	6.1	7.4	6.7	35.3	36.4	35.9
	OVIS2.5-2B [27]	63.1	50.7	56.2	10.4	10.1	10.3	42.6	35.4	38.7
	DeepSeek-VL2-Small [42]	52.1	73.3	60.9	12.5	18.5	14.9	39.1	55.4	45.9
	Qwen2.5-VL-7B [1]	55.4	75.8	64.0	10.6	15.5	12.6	39.5	54.2	45.7
	Qwen2.5-VL-3B [1]	55.7	77.2	64.7	12.7	18.3	15.0	41.0	56.8	47.6
	SEED1.5-VL [13]	65.3	**78.6**	71.3	12.7	16.4	14.3	46.8	56.9	51.4
	Qwen3-VL-4B [47]	58.6	70.5	63.7	13.7	16.1	14.8	43.4	52.0	47.1
	Qwen3-VL-8B [47]	58.4	71.4	64.2	14.1	19.1	16.2	43.4	53.8	48.1
	Rex-Omni [16]	68.1	76.3	72.0	14.5	17.5	15.9	49.8	56.5	52.9
	HiLoc	**69.6**	77.0	**73.1**	**20.1**	**22.8**	**21.3**	**53.4**	**59.6**	**56.3**

3.4 Data Engine for HiRef-500K

Existing referring expression datasets [18, 29, 52] contain almost no negative examples and use only simple phrases, limiting both *rejection* learning and *complex referring* ability. To address this, we build a data engine on detection datasets (Figure 4): for each $(\texttt{image_id}, \texttt{category_id})$ pair, all ground-truth annotations are fed to Qwen3-VL-32B-Instruct [47], which generates expressions of five types: attribute, state, relation, set logic, and *nonexist*. The nonexist type describes objects *absent* from the image, providing direct supervision for rejection mechanism. The engine yields $\sim$500K annotated expressions in total, which we term **HiRef-500K**; full details are in Appendix F.

4 Experiments

4.1 Experimental Architecture and Setup

Base MLLM. We instantiate HiLoc on top of Qwen3-VL-4B-Instruct [47]. We extend the tokenizer with 64 discrete index tokens ($\langle 0\rangle$–$\langle 63\rangle$) and 11 structured tags.

Training data and baselines. Our SFT data spans three tiers: (1) general detection: APTv2 [48], BDD100K [51], COCO [23], Objects365-v2 [37], OpenImages-v7 [19], LVIS [14], V3Det [40], EgoObjects [57], HumanParts [22], ImageNet-Part [15], PACO [32], DeepFashion2 [11]), (2) referring & grounding: RefCOCO/+/g [18, 52], Ref-L4 [3], and our HiRef-500K dataset, and (3) domain-specific imagery: VisDrone [10], DOTA-v2 [43], FAIR1M [38]), totaling approximately 4.4M SFT samples. For GRPO post-training, we use a subset of approximately 80K samples. Baselines include closed-set detectors [2, 8, 24, 33, 55, 58], open-vocabulary detectors [9, 21, 25, 30], and MLLM-based methods [1, 13, 16, 27, 28, 42, 44, 47, 49, 54].

Evaluation Metric and Settings. We follow the evaluation protocol of Rex-Omni [16]. Since most MLLMs perform poorly when querying multiple phrases in a single conversation, all methods

Table 2: Performance Comparison on LVIS Dataset for Long-tailed Object Detection.

Type	Method	LVIS [14]								
		R@IoU$_{0.5}$	P@IoU$_{0.5}$	$F1$@IoU$_{0.5}$	R@IoU$_{0.95}$	P@IoU$_{0.95}$	$F1$@IoU$_{0.95}$	R@mIoU	P@mIoU	$F1$@mIoU
Open-set	Grounding DINO-Swin-T [25]	39.1	61.2	47.7	16.9	34.8	22.7	31.9	49.6	38.8
MLLM	DeepSeek-VL2-Tiny [42]	22.4	55.7	32.0	7.2	25.7	11.2	14.2	41.9	21.2
	MiMo-VL-7B [44]	43.3	57.8	49.5	6.5	13.7	8.8	25.4	41.1	31.4
	OVIS2.5-2B [27]	53.2	55.6	54.4	13.3	19.4	15.8	34.0	41.6	37.4
	OVIS2.5-9B [27]	52.2	54.1	53.1	11.2	19.9	14.4	32.2	40.1	35.8
	Qwen2.5-VL-7B [1]	47.1	74.3	57.7	12.7	29.0	17.6	32.0	54.3	40.2
	Qwen2.5-VL-3B [1]	44.8	73.9	55.8	14.3	29.8	19.3	31.9	54.6	40.3
	DeepSeek-VL2-Small [42]	46.1	72.2	56.2	15.8	31.2	21.0	33.7	55.0	41.8
	SEED1.5-VL [13]	54.7	**82.0**	65.6	15.0	28.1	19.5	38.5	59.3	46.7
	Qwen3-VL-4B [47]	49.3	81.4	61.4	15.7	32.8	21.2	35.7	**61.7**	45.3
	Qwen3-VL-8B [47]	46.5	80.3	58.9	14.7	32.3	20.2	33.4	60.2	43.0
	Rex-Omni [16]	54.7	78.1	64.3	16.5	27.6	20.7	39.6	57.5	46.9
	HiLoc [16]	**58.3**	80.5	**67.6**	**19.3**	**34.8**	**24.8**	**42.9**	61.1	**50.4**

Table 3: Performance Comparison on VisDrone and RefCOCOg Benchmarks.

Type	Method	VisDrone [17]			RefCOCOg val [52]			RefCOCOg test [18]		
		$F1$@IoU$_{0.5}$	$F1$@IoU$_{0.95}$	$F1$@mIoU	$F1$@IoU$_{0.5}$	$F1$@IoU$_{0.95}$	$F1$@mIoU	$F1$@IoU$_{0.5}$	$F1$@IoU$_{0.95}$	$F1$@mIoU
Open-set	Grounding DINO-Swin-T [25]	28.0	16.5	25.2	52.9	20.9	45.9	53.8	22.9	46.8
MLLM	DeepSeek-VL2-Tiny [42]	4.3	0.1	1.8	67.4	16.1	50.5	69.3	16.9	52.1
	OVIS2.5-2B [27]	21.0	0.1	9.2	87.4	29.3	73.4	87.6	30.5	73.8
	OVIS2.5-9B [27]	15.8	0.1	6.5	88.8	23.5	72.1	88.7	24.2	72.6
	Qwen2.5-VL-3B [1]	31.5	1.9	20.4	83.5	30.7	69.2	83.8	31.8	70.1
	MiMo-VL-7B []	27.7	0.3	14.3	84.9	14.4	65.3	84.6	14.9	65.5
	Qwen2.5-VL-7B [1]	34.5	1.6	21.7	86.2	27.2	70.0	85.7	28.4	70.4
	DeepSeek-VL2-Small [42]	35.8	1.7	23.3	**92.4**	45.6	81.4	**91.8**	47.0	**81.6**
	SEED1.5-VL [13]	55.9	0.6	27.4	84.7	30.9	71.9	85.2	32.1	73.2
	Qwen3VL-4B [47]	26.8	1.1	17.3	87.5	35.3	74.2	88.3	36.7	75.6
	Qwen3VL-8B [47]	25.5	1.1	16.3	88.3	36.7	75.6	87.5	35.5	74.8
	Rex-Omni [16]	**61.6**	1.5	35.8	86.6	35.3	73.6	86.8	36.6	74.3
	HiLoc	59.9	**3.9**	**37.9**	89.0	**55.6**	**81.5**	88.8	**53.9**	80.6

except Rex-Omni, SEED1.5-VL, and our HiLoc query one phrase at a time; these three methods query all phrases in a single conversation turn. As MLLMs lack reliable confidence scores, we use macro-averaged $F1$ (with IoU at 0.5, 0.95, and mIoU) as the primary metric; OmniLabel is measured by AP-d and its positive/size splits; D^3 reports Full, Presence, and Absence AP, where Absence AP directly evaluates rejection capability. Full metric definitions are in Appendix B.

Implementation details. In full-parameter SFT, the vision encoder uses a learning rate of 10^{-6}, while the projection layer and LLM share a higher rate of 10^{-5}. GRPO post-training uses G=8 candidates per prompt, KL coefficient β=0.02. Complete hyperparameters are in Appendix B.

4.2 Main Results

General detection (COCO). As shown in Table 1, HiLoc achieves the best $F1$@IoU$_{0.5}$ (73.1) among MLLM methods, surpassing Rex-Omni by 1.1 points. At the stricter IoU$_{0.95}$, HiLoc obtains 21.3, outperforming Rex-Omni by 5.4 points, demonstrating the precision advantage of HiLoc.

Large-vocabulary detection (LVIS). Table 2 shows that HiLoc achieves the highest $F1$@IoU$_{0.5}$ of 67.6, outperforming Rex-Omni by 3.3 and SEED1.5-VL by 2.0 points. At IoU$_{0.95}$, HiLoc reaches 24.8 vs. Rex-Omni's 20.7, confirming HiLoc's precision benefit in large-vocabulary settings.

Referring expression comprehension (RefCOCOg). Table 3 reports results on both val and test splits. At IoU$_{0.95}$, HiLoc achieves 55.6/53.9, surpassing DeepSeek-VL2-Small by 10.0/6.9 points and Rex-Omni by 20.3/17.3 points, confirming the capability for complex understanding and localization.

Aerial remote sensing detection (VisDrone). Table 3 evaluates the performance on detecting small, dense objects. HiLoc achieves the best $F1$@mIoU of 37.9 (+2.1 over Rex-Omni) and $F1$@IoU$_{0.95}$ of 3.9 (2.6× Rex-Omni's 1.5), demonstrating that the coarse-to-fine decomposition effectively narrows search regions in large aerial images while providing fine-grained precision for small targets.

Description-based detection (OmniLabel & D^3). Table 4 evaluates description-based object detection with negative samples. On OmniLabel, it achieves AP-d of 23.0, surpassing Rex-Omni by 11.9. On D^3, HiLoc obtains Full AP of 53.1, far exceeding others, validating strong rejection ability.

Table 4: Performance comparison on OmniLabel and D^3 benchmarks, which contain negative samples. AP-d represents Average Precision for descriptions

Type	Method	OmniLabel [36]					D^3 [45]		
		AP-d	AP-dPositive	AP-dSmall	AP-dMedium	AP-dLarge	Full	Presence	Absence
Specialists	RegionCLIP [56]	2.6	3.2	3.6	2.7	2.3	–	–	–
	OFA-DOD [46]	–	–	–	–	–	21.6	23.7	15.4
	Grounding DINO-Swin-T [25]	–	–	–	–	–	20.7	20.1	22.5
	OWLv2 [30]	21.9	26.9	32.9	20.5	14.4	22.8	22.1	24.7
	GLIP-T [21]	16.4	25.8	29.4	14.8	8.2	19.1	18.3	21.5
	FIBER-B [9]	22.3	34.8	**38.6**	19.5	12.4	22.7	21.5	26.0
Generalists	*MLLM with a larger parameter scale*								
	InternVL2-76B [6]	15.3	19.3	21.9	15.2	11.3	25.3	25.7	23.5
	InternVL2-8B [6]	4.5	5.2	6.2	4.2	3.9	9.8	11.0	6.2
	InternVL3-8B [6]	4.0	15.1	4.8	4.0	4.2	14.3	14.3	14.3
	Griffon(13B) [54]	7.2	12.7	12.5	6.3	4.5	12.3	12.4	12.2
	Groma(7B) [28]	15.6	27.2	21.4	15.6	11.3	16.0	15.9	16.3
	ROD-MLLM(7B) [49]	**25.3**	30.9	31.8	**24.5**	**21.0**	29.7	30.0	28.7
	Qwen3VL-4B [47]	12.6	27.6	21.1	11.9	10.0	28.8	29.9	25.4
	Qwen3VL-8B [47]	9.24	19.59	10.90	9.12	10.71	30.6	31.5	27.9
	Rex-Omni (3B) [47]	12.9	33.6	20.0	12.2	11.0	21.5	22.1	19.8
	HiLoc (4B)	23.0	**41.4**	38.5	21.4	16.8	**53.1**	**55.3**	**46.6**



Figure 5: Qualitative comparison of HiLoc against baseline methods. The images are obtained from LVIS[14], VisDrone [10], RefCOCOg [52] and D-Cube [45].

Qualitative Comparison. Figure 5 and Appendix G present visual comparisons between HiLoc and baseline methods across diverse scenarios. HiLoc produces tighter bounding boxes with fewer missed or spurious detections, particularly for small, occluded, or description-specified objects.

4.3 Ablations

We conduct a series of ablation studies to dissect the contribution of each key design in HiLoc, with additional studies on the effectiveness of HiRef-500K (Appendix C.1) and the sensitivity to Grid/Cell resolution M, N (Appendix C.2) deferred to the appendix.

Performance scaling with IoU and object size. HiLoc's relative F1 gain over flat-coordinate baselines (Rex-Omni, Qwen3-VL-4B) increases steadily as the IoU threshold τ tightens (see Figure 6 (a,b)). Figure 6(c) further shows that HiLoc's advantage is even clearer on small object detection.

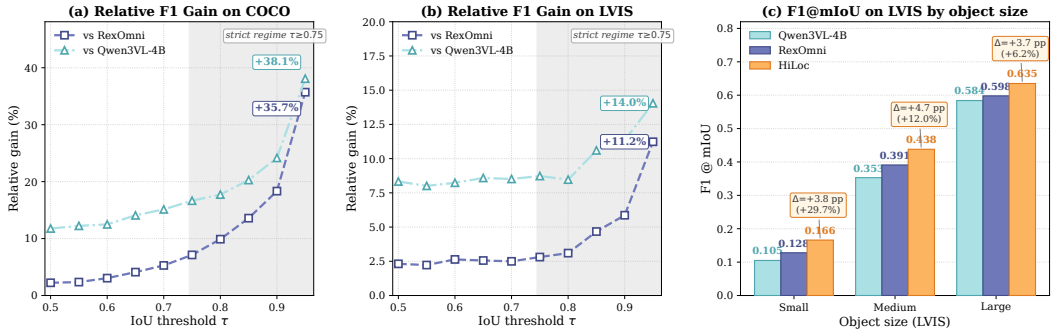


Figure 6: **Performance scaling with IoU and object size.** HiLoc's performance advantage is more pronounced at higher IoU thresholds and on smaller objects. **(a,b)** Relative F1 gain of HiLoc over flat-coordinate baselines (RexOmni, Qwen3VL-4B) as the IoU threshold τ tightens on COCO and LVIS dataset. **(c)** F1@mIoU on LVIS by object size. The numbers inside the orange box indicate the absolute gain and the relative gain percentage compared to Rex-Omni.

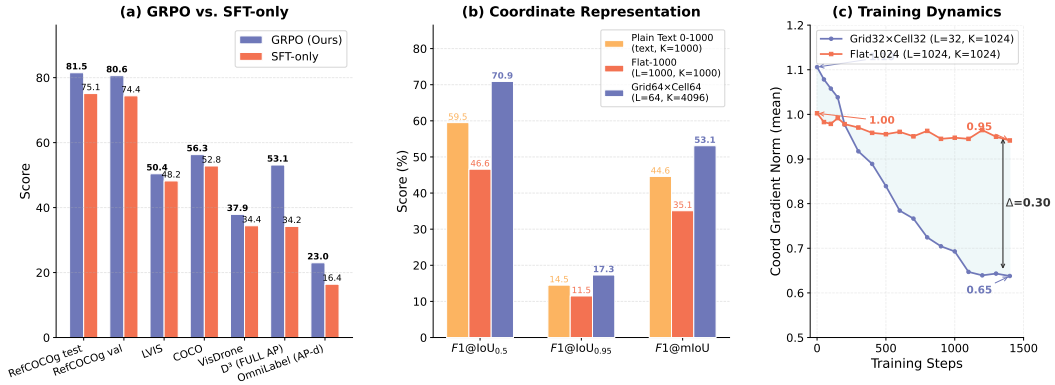


Figure 7: **(a) Effect of GRPO Training. (b) Coordinate Representation Comparison**: Let L denotes the number of special tokens to represent coordinates, and K denotes the resolution per axis. We compare three representation methods's performance on COCO dataset : *Plain Text 0-1000* ($K = 1000$, Qwen3VL-4B's format), *Flat-1000* ($L = 1000, K = 1000$, Rex-Omni's Format), and *Grid64×Cell64* ($L = 64, K = 4096$, HiLoc's Format). Under the same training conditions, HiLoc's representation method is more effective. **(c) Training dynamics.** At the same resolution, *Grid32×Cell32*'s ($L = 32, K = 1024$, HiLoc's Format) mean coordinate gradient norm declines much more markedly than Flat-1024's ($L = 1024, K = 1024$, Rex-Omni's Format), indicating that HiLoc enables more stable and easier optimization.

Effect of GRPO Training. As shown in Figure 7(a), adding GRPO training after SFT brings consistent improvements across all six benchmarks, effectively enhancing the model's performance.

Effect of Coordinate Representation. We train Qwen3VL-4B on the same general detection datasets (COCO / LVIS training set, etc.), differing only in the coordinate representation format. Specifically, we compare: Plain Text 0–1000 (following the original format of Qwen3VL-4B), Flat-1000 (using 1000 special tokens, following Rex-Omni's format), and Grid64×Cell64 (HiLoc with $M, N = 64$). As shown in Figure 7(b), Grid64×Cell64 achieves the highest F1 across all IoU thresholds. In contrast, HiLoc's representation introduces far fewer learnable special tokens (only $L = 64$ special tokens, compared to $L = 1000$ in Flat-1000), supports a much higher resolution per axis (e.g., $K = 64 \times 64 = 4096$ discrete locations per axis, compared to $K = 1000$ in Flat-1000).

Training Dynamics. Under identical training settings and the same effective resolution ($K = 1024$ per axis), we compare Flat-1024 (using 1024 special tokens, following Rex-Omni's format) and Grid32×Cell32 (our HiLoc coordinate representation, with $M, N = 32$). Figure 7(c) tracks the mean coordinate gradient norm during the training. Grid32×Cell32 exhibits a steadily decreasing norm

throughout training, whereas Flat-1024 plateaus early. This corroborates Proposition 3.2: the smaller per-step classification space prevents gradient dilution, sustaining effective optimization.

4.4 Additional Controlled Experiments

Resolution and vocabulary budget. Figure 7(b) compares practical coordinate formats under the same Qwen3-VL-4B backbone and training settings, but with different resolutions. The additional four-variant study in Table 5 addresses this confound. At $K = 1024$, Grid32×Cell32 exceeds Flat-1024 by 10.3, 1.2, and 6.8 F1 points while using 32 rather than 1,024 coordinate tokens. At $L = 64$, Grid64×Cell64 exceeds Flat-64 by 17.8, 10.1, and 17.1 points, while also increasing resolution from 64 to 4,096. The equal-resolution result supports an advantage beyond finer discretization; the equal-budget result illustrates the vocabulary–resolution trade-off. These detection-only studies use a different training mixture from the full model.

Table 5: Controlled COCO validation comparison using the same Qwen3-VL-4B backbone, detection-related training data, and optimization settings. L counts coordinate tokens; K is per-axis resolution. Scores are F1 in percent.

Representation	L	K	IoU 0.5	IoU 0.95	mean IoU
Flat-64	64	64	54.5	8.6	37.7
Flat-1024	1024	1024	56.2	13.4	41.8
Grid32×Cell32	32	1024	66.5	14.6	48.6
Grid64×Cell64	64	4096	72.3	18.7	54.8

Matched data and reward on D^3. Table 6 compares Flat-1000 and HiLoc with the same backbone, phrase/box output structure, rejection token, HiRef data, and SFT/GRPO settings. HiLoc leads by 8.4 AP after SFT and 10.5 AP after GRPO. GRPO improves both representations (16.8 and 18.9 points), so the gains are not attributed to representation alone. This experiment controls data and reward but not effective resolution. Appendix C.1 separately evaluates the addition of HiRef.

Table 6: Matched-data/reward comparison on D^3 (Full AP, percent). Backbone, phrase/box grammar, `<no_box>`, HiRef data, and stage-specific training settings are shared. Resolution differs: $K = 1000$ for Flat and $K = 4096$ for HiLoc.

Representation	SFT	SFT + GRPO
Flat-1000	25.8	42.6
HiLoc (Grid64×Cell64)	34.2	53.1

Token-sharing results are provided in Appendix I.1.

5 Conclusion

In this work, we present HiLoc, which reformulates spatial localization in MLLM as *coarse-to-fine hierarchical index prediction* with a small set of learnable special tokens. HiLoc achieves quadratic resolution scaling with better training stability and lower quantization error. We also contribute HiRef-500K, a dataset with negative examples and complex phrases to improve rejection and grounding ability. Experiments show HiLoc excels at high IoU, small objects, and complex referring, offering a strong alternative to flat coordinate representations for MLLM-based localization.

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A Proofs of Theoretical Results

We provide complete proofs for the theoretical results stated in Section 3.2.

Clarification. The original analyses below are retained; Appendix H supplies corrected decoder-dependent bounds and qualifications on training stability and error locality. Throughout, we consider an image of size $W \times H$ partitioned into an $M \times M$ coarse grid with $N \times N$ fine cells per grid, giving effective resolution $K = MN$ per axis. Grid and Cell indices share the same token vocabulary $\langle 0 \rangle, \langle 1 \rangle, \ldots, \langle \max(M, N)-1 \rangle$.

A.1 Proof of Proposition 3.1: Quantization Error Bound

Proof. Under HiLoc, the decoded x-coordinate is $\hat{P}_w = G_i \cdot G_w + C_u \cdot C_w$, where $G_w = W/M$ is the grid width and $C_w = W/(MN)$ is the cell width. The true coordinate satisfies $P_w \in [\hat{P}_w,\ \hat{P}_w + C_w)$, so the maximum quantization error is

$$\epsilon_{\text{Hi}} = \frac{C_w}{2} = \frac{W}{2MN} = \frac{W}{2K}. \tag{6}$$

For flat single-level quantization with K bins of width W/K, the identical argument gives $\epsilon_{\text{flat}} = W/(2K)$. Thus, at equal effective resolution $K = MN$, both methods achieve the same quantization precision.

The key difference lies in the *per-step classification space*. HiLoc decomposes each coordinate into two predictions over M and N classes respectively, sharing the token set $\{\langle 0 \rangle, \ldots, \langle \max(M, N)-1 \rangle\}$, while flat quantization requires a single prediction over $K = MN$ classes. When $M = N = 64$, HiLoc needs only 64 token types to achieve $K = 4096$ effective resolution, a compression ratio of $64\times$. □

Remark: vocabulary-budget analysis. Under a fixed token vocabulary budget L, flat quantization achieves resolution $K_{\text{flat}} = L$, while HiLoc can set $M = N = L$ to achieve $K_{\text{Hi}} = L^2$. The quantization error ratio is therefore

$$\frac{\epsilon_{\text{flat}}}{\epsilon_{\text{Hi}}} = \frac{W/(2L)}{W/(2L^2)} = L, \tag{7}$$

i.e., HiLoc achieves $L\times$ finer precision with the same vocabulary size. For example, with $L = 64$ tokens, flat quantization resolves 64 bins ($\epsilon = W/128$), while HiLoc resolves $64^2 = 4096$ bins ($\epsilon = W/8192$).

A.2 Proof of Proposition 3.2: Gradient Variance and Training Stability

We first establish a supporting lemma on CE gradient structure, then prove the training stability result.

CE gradient analysis. Consider predicting a spatial index via softmax cross-entropy over a vocabulary of size L. Let $p^* = p_\theta(y^* \mid \cdot)$ denote the model's predicted probability for the correct class y^*. The gradient of the CE loss with respect to the logit z_{y^*} of the correct class is

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{y^*}} = p^* - 1, \tag{8}$$

and for any incorrect class $j \neq y^*$,

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_j} = p_j. \tag{9}$$

Under uniform initialization where $p_j \approx 1/V$ for all j, the squared gradient norm over incorrect classes is

$$\sum_{j \neq y^*} \left(\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_j} \right)^2 \approx (L-1) \cdot \frac{1}{L^2} = \frac{L-1}{L^2} \approx \frac{1}{L}. \tag{10}$$

Proof of Proposition 3.2. Under uniform initialization, the entropy of the softmax distribution is $H = \log L$. For flat quantization, $H_{\text{flat}} = \log K = \log(MN)$, while for each HiLoc prediction step, $H_{\text{Hi}} = \log(\max(M, N))$. The initial cross-entropy loss satisfies

$$\mathcal{L}_{\text{CE}}^{\text{init}}(\text{flat}) = \log(MN) = \log M + \log N, \quad \mathcal{L}_{\text{CE}}^{\text{init}}(\text{Hi, per step}) = \log(\max(M, N)). \tag{11}$$

The *effective gradient signal-to-noise ratio* (SNR) at initialization, defined as the ratio of the correct-class gradient magnitude to the total incorrect-class gradient norm, scales as

$$\text{SNR} \propto \frac{|p^* - 1|}{\sqrt{\sum_{j \neq y^*} p_j^2}} = \frac{1 - 1/L}{\sqrt{(L-1)/L^2}} = \sqrt{L-1} \approx \sqrt{L}. \tag{12}$$

While a higher SNR means the correct-class signal is more distinguishable, and $L_{\text{Hi}} < L_{\text{flat}}$ implies a lower per-step SNR for HiLoc, the critical advantage is that each step's classification problem is *well-conditioned*: the model distinguishes among $\max(M, N)$ spatially coherent regions rather than navigating a $K = MN$-way classification landscape.

Concretely, the number of "distractor" logits that compete with the correct class is $\max(M, N) - 1$ per step for HiLoc versus $MN - 1$ for flat methods. Under gradient descent, convergence rate for L-way classification scales as $O(L \log L)$ iterations to reach ϵ-accuracy, so HiLoc's per-step convergence is $O(\max(M, N) \log \max(M, N))$ versus $O(MN \log(MN))$ for flat methods. □

A.3 Proof of Proposition 3.3: Error Propagation Robustness

Proof. We decompose the expected localization error by conditioning on which hierarchical level(s) contain prediction errors.

Case 1: Grid error (probability p_g). When the Grid prediction is wrong, the decoded point falls in an incorrect coarse region. The displacement is at most one grid cell width $G_w = W/M$ (for an off-by-one error; larger errors are exponentially less likely under CE training, as the softmax probability mass concentrates on nearby indices).

Case 2: Grid correct, Cell error (probability $(1 - p_g)\, p_c$). When the Grid is correct but the Cell prediction is wrong, the point falls in the correct coarse region but the wrong fine cell. The displacement is at most one cell width $C_w = W/(MN)$.

Case 3: Both correct (probability $(1 - p_g)(1 - p_c)$). Only quantization error remains: $W/(2MN)$.

Summing these contributions:

$$\mathbb{E}\big[|\hat{P}_w - P_w|\big]_{\text{Hi}} \leq p_g \cdot \frac{W}{M} + (1 - p_g)\, p_c \cdot \frac{W}{MN} + (1 - p_g)(1 - p_c) \cdot \frac{W}{2MN}. \tag{13}$$

□

Detailed comparison with flat methods. For flat coordinate generation that produces D digit tokens sequentially (e.g., $D = 3$ digits for 0–999), an error in the d-th most significant digit causes a displacement of $O(W \cdot 10^{-d})$. Let p_d denote the error probability for the d-th digit. The expected error satisfies

$$\mathbb{E}\big[|\hat{P}_w - P_w|\big]_{\text{flat}} = \sum_{d=1}^{D} p_d \cdot \frac{W}{10^d} \;\geq\; p_1 \cdot \frac{W}{10}. \tag{14}$$

Comparing the two:

- HiLoc's worst-case single-level error (Grid mistake) is W/M. For $M = 64$: $W/64 \approx 0.016W$.
- Flat method's worst-case single-digit error (leading digit) is $W/10 = 0.1W$.
- HiLoc's error is *structured*: Grid errors produce interpretable region-level shifts, while Cell errors produce bounded local perturbations, enabling graceful degradation.
- In flat methods, a single wrong digit at any position can cause unpredictable jumps, and the error magnitude depends on the digit's positional significance in a way that is opaque to the model.

B Experimental Setup Details

We provide the complete experimental setup details for Section 4.1.

B.1 Base MLLM and Tokenizer

We instantiate HiLoc on top of Qwen3-VL-4B-Instruct [47], a strong MLLM backbone following the Qwen-style vision-language interface. The model consists of (i) a ViT-based vision encoder that converts the input image into a sequence of visual tokens, (ii) a multimodal projector (MLP) that maps visual features to the language embedding space, and (iii) a causal language model (LLM) that performs autoregressive decoding conditioned on both text tokens and projected visual tokens.

We extend the base tokenizer with the discrete index tokens $\langle 0\rangle, \langle 1\rangle, \ldots, \langle 63\rangle$ and the structured tags (, , <box>, </box>, <point>, </point>, <grid>, </grid>, <cell>, </cell>, <no_box>). The default configuration uses $M{=}64$ coarse grid divisions and $N{=}64$ fine cell divisions, yielding an effective resolution of $K{=}4096$ per axis with only 64 index tokens. Additional qualitative comparisons of the resulting detection and grounding performance are provided in Appendix G (Figures 11 and 12). The coordinate space operates on the resized image dimensions (H', W') computed by the model's `smart_resize` function with a patch factor of 32, ensuring alignment between the vision encoder's spatial resolution and the coordinate system.

B.2 Training Datasets

For SFT, we curate a large-scale multi-source detection dataset spanning diverse visual domains:

- **General detection**: APTv2 [48], BDD100K [51], COCO [23], Objects365-v2 [37], OpenImages-v7 [19], LVIS [14], V3Det [40], EgoObjects [57], HumanParts [22], ImageNet-Part [15], PACO [32] and DeepFashion2 [11].
- **Referring and grounding**: RefCOCO/+/g [18, 52], Ref-L4 [3], and HiRef-500K (Section 3.4).
- **Domain-specific**: VisDrone [10] , DOTA-v2 [43] and FAIR1M [38].

All annotations are converted to the HiLoc's format using a unified conversion pipeline, yielding approximately 4.4M training samples in total for SFT. For GRPO post-training, we use a subset of approximately 80K samples from the SFT data.

System prompt and data format. During training, each sample is prepended with the following system prompt that defines the GridCell Format specification:

```
You are a grounding formatter. You must follow the instruction strictly.
We define a special output format named "GridCell Format".
When the user asks you to use GridCell Format, you MUST output results in GridCell
    Format.

Tag semantics (meanings):
- <p>phrase</p>: a category/phrase group. The text inside <p>...</p> is the category
    name.
- <box>...</box>: one detected instance (one bounding box) for the nearest preceding
    <p>phrase</p>.
  A <box> contains exactly TWO points: top-left and bottom-right.
- <point>...</point>: one point describing a corner of the box.
- <grid>Gi Gj</grid>: the coarse grid index (Gi, Gj) for this point.
- <cell>Cu Cv</cell>: the fine cell index (Cu, Cv) inside the chosen grid.
- <no_box>: means there is NO instance for that phrase/category.
- <k>: numeric index token. Indices MUST be expressed using numeric tokens like <0>,
    <1>, ...

GridCell indexing definition (math):
Assume the input image has width W and height H.
In this dataset, we FIX M = 64 (Grid) and N = 64 (Cell).
Therefore valid indices are Gi,Gj in [0,63] and Cu,Cv in [0,63], represented as
    tokens <0>..<63>.

GridCell Format rules:
1) Output MUST be in the tag format only.
2) For each category, output: <p>category</p> followed by either <no_box> or one/more
     <box>...</box>.
```

```
3) Each <box> MUST contain two points (top-left and bottom-right):
   <point><grid><Gi><Gj></grid><cell><Cu><Cv></cell></point>,<point><grid><Gi><Gj></
       grid><cell><Cu><Cv></cell></point>
4) Numeric indices MUST be expressed as tokens <0>..<63>.
5) Do not output any extra words outside the tags.
```

The corresponding user prompt follows a fixed template:

```
Use GridCell Format to output grounding results. Categories: phrase1, phrase2, ...
    phraseN. If none, output <no_box>.
```

where `phrase1, phrase2, ..., phraseN` are replaced by the actual category or referring-expression queries for each sample.

Each training sample consists of an image paired with one or more phrases to be localized, with a maximum of 10 phrases per image. For each phrase, the model is expected to output the corresponding bounding boxes in GridCell Format, or `<no_box>` if the described object is absent from the image.

B.3 Baselines

We compare HiLoc against three categories of methods:

- **Closed-set detectors** (6 methods): Faster R-CNN [33], DETR [2], DyHead [8], DAB-DETR [24], Deformable-DETR [58], and DINO [55].
- **Open-vocabulary detectors** (4 methods): Grounding DINO [25], GLIP [21], OWLv2 [30], and FIBER [9].
- **MLLM-based localization** (13 methods): Qwen2.5-VL (3B/7B) [1], Qwen3-VL (4B/8B) [47], DeepSeek-VL2 (Tiny/Small) [42], SEED1.5-VL [13], OVIS2.5 (2B/9B) [27], MiMo-VL-7B [44], ROD-MLLM [49], Groma [28], Griffon [54], and Rex-Omni [16].

B.4 Evaluation Benchmarks and Metrics

Metric rationale. Following Rex-Omni [16], we adopt Recall (R), Precision (P), and $F1$ score. For each category, R and P are computed from predicted and ground-truth boxes, then macro-averaged; $F1$ is their harmonic mean. Following COCO, Intersection over Union (IoU) is evaluated at thresholds from 0.5 to 0.95 (step 0.05), and we report results at IoU=0.5, IoU=0.95, and the mean across thresholds (mIoU).

Benchmark-specific metrics. We evaluate on six benchmarks:

- **COCO** [23] and **LVIS** [14]: Macro-averaged R, P, and $F1$ at IoU 0.5, 0.95, and mIoU.
- **OmniLabel** [36]: AP-d (description-conditioned AP), which evaluates how well a model localizes objects given free-form text descriptions. We report AP-d overall, AP-dPositive (P, restricted to descriptions with at least one matching object), and AP-d by object size (S/M/L for Small/Medium/Large).
- D^3 [45]: Full AP, Presence AP (Pres, descriptions with matching objects), and Absence AP (Abs, descriptions with no matching objects). Absence AP directly evaluates the model's rejection capability—its ability to correctly predict no box when the described object is absent.
- **RefCOCOg** [52]: $F1$ at IoU 0.5, 0.95, and mIoU on both val and test splits.
- **VisDrone** [10]: $F1$ at IoU 0.5, 0.95, and mIoU to evaluate performance on small, dense aerial objects.

B.5 Implementation Details

SFT stage. We train with full-parameter updates (vision encoder, projector, and LLM) using DeepSpeed ZeRO-3. The learning rate is 1×10^{-5} for the LLM, 1×10^{-5} for the projector, and 1×10^{-6} for the vision encoder, with a cosine schedule and 3% linear warmup. We use a per-device batch size of 8 with gradient accumulation of 8 steps, weight decay of 0.01. The maximum sequence

length is 4096 tokens. Images are resized with dynamic resolution, with pixel counts ranging from $16{,}384$ to $1{,}048{,}576$. Training uses BF16 mixed precision with gradient checkpointing.

GRPO stage. Starting from the SFT checkpoint, we run GRPO with a learning rate of 5×10^{-6}, and KL regularization coefficient β=0.02. For each prompt, we sample G=8 candidate completions. The maximum prompt and completion lengths are both 4096 tokens. The detection reward uses F_β with β=0.8, IoU threshold τ=0.5, miss penalty λ_{fn}=0.2, false-positive penalty γ_{fp}=0.3, correct-rejection reward r_0=0.8, and false-prediction penalty α=0.3. The reward weights are λ_{det}=1.0 and λ_{fmt}=0.1.

Compute resources. All experiments are conducted on a single multi-GPU server equipped with 8 high-end GPUs. Training uses BF16 mixed precision together with DeepSpeed ZeRO-3 and gradient checkpointing to fit the full-parameter updates within device memory. Under this configuration, a complete training run (SFT followed by GRPO) takes approximately one week.

C Additional Ablation Studies

C.1 HiRef-500K Dataset Effectiveness

We validate HiRef-500K's effectiveness by comparing SFT performance with and without this dataset. Two HiLoc variants are trained: one with the full 4.4M-sample dataset including HiRef-500K, and another excluding it. All other training configurations remain identical.

Table 7: Ablation study on HiRef-500K dataset effectiveness.

Benchmark	Metric	w/ HiRef-500K	w/o HiRef-500K
RefCOCOg	IoU@0.5	0.827	0.813
	IoU@0.95	0.489	0.467
	mIoU	0.751	0.734
D^3	FULL	0.342	0.108
	Pres	0.346	0.125
	Abs	0.270	0.058

As shown in Table 7, results show consistent improvements across all metrics when including HiRef-500K: RefCOCOg mIoU increases by 0.017, while D^3 metrics show substantial gains (FULL AP: +0.234, Absence AP: +0.212). The significant Absence AP improvement (0.270 vs 0.058) demonstrates enhanced rejection capability for nonexistent objects. These findings confirm HiRef-500K's value in providing diverse referring expressions for robust localization training.

C.2 Grid and Cell Resolution

The default configuration in the main paper fixes the coarse Grid and fine Cell to $M{=}N{=}64$, yielding an effective spatial resolution of $K{=}MN{=}4096$ per axis. We here justify this choice empirically by ablating the Grid/Cell size over $\{16, 32, 64, 128\}$ (with $M{=}N$ throughout) and evaluating the detection $F1$ curve across IoU thresholds $\{0.50, 0.55, \ldots, 0.95\}$.

We fine-tune from the same Qwen3-VL-4B base for all four configurations on the COCO single-annotation subset described above (only images with exactly one GT box are retained for both the train and val splits), using identical optimizer, batch size and schedule. Checkpoints are logged every 500 steps up to 4200 steps and evaluated on the held-out single-annotation val split; since every image has exactly one target, this setting gives a clean measurement insensitive to instance-matching noise.

Figure 8 shows three representative snapshots (early / intermediate / converged). Two trends emerge consistently. First, at checkpoint-500 the highest-resolution setting $M{=}N{=}128$ has not yet fit the hierarchical index space well and is actually outperformed by $M{=}N{=}64$ ($F1_\text{mIoU}$ 0.515 vs. 0.476), matching the gradient-stability argument in Proposition 3.2: a larger per-step classification space ($\max(M, N){=}128$ vs. 64) has a lower early-stage signal-to-noise ratio and needs more iterations to become well-conditioned. Second, at checkpoint-4200 the two low-resolution settings saturate clearly below the others ($F1_\text{mIoU}{=}0.633$ for 16, 0.711 for 32), consistent with their larger quantization error $W/(2MN)$ (Proposition 3.1), while 64 and 128 converge to almost identical quality (0.773 vs. 0.788; $\Delta{=}0.015$) – effective resolution has already saturated at $K{=}64{\times}64{=}4096$ per axis.

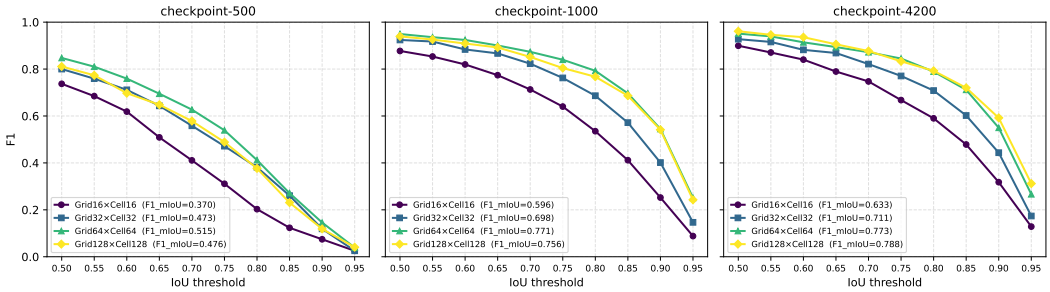


Figure 8: Ablation on the resolution of Grid and Cell (M, N), trained and evaluated on the COCO single-annotation subset. Each subplot fixes one training checkpoint (500 / 1000 / 4200 steps) and plots $F1$ vs. IoU threshold for four configurations ($M{=}N \in \{16, 32, 64, 128\}$).

Combining the two observations, $M{=}N{=}64$ gives the best trade-off between training stability and sample efficiency (strictly faster early convergence than 128), final accuracy (within 0.015 mIoU of the best 128 setting and clearly ahead of 16/32), and vocabulary cost (only 64 numeric index tokens for an effective resolution of 4096 per axis). We therefore adopt $M{=}N{=}64$ as the default configuration throughout the main experiments.

D Comparison with Grid–Offset Tokenization (GETok)

A concurrent line of work, GETok [35], also introduces discrete spatial tokens for MLLM-based localization: it first uses *grid tokens* to partition the image plane into structured spatial anchors, and then exploits *offset tokens* to enable precise and iterative refinement of localization predictions. Although both HiLoc and GETok replace flat coordinate generation with discrete spatial tokens, they differ fundamentally in two aspects that we highlight below.

Semantics of special tokens. GETok employs two *semantically heterogeneous* families of tokens: (i) *grid tokens* of the form $\langle \mathrm{GRID}_{i,j} \rangle$, which enumerate all $n \times n$ anchors over the image as independent atomic symbols, and (ii) *offset tokens* of the form $\langle \mathrm{OFF}_{\delta u, \delta v} \rangle$ taking values in $\{-1, 0, +1\}$, which encode *relative displacement vectors* indicating how to move from the current anchor, together with a $\langle \mathrm{DELETE} \rangle$ token for undoing a point. The resulting vocabulary is a hybrid of "absolute anchors + relative vectors". In contrast, HiLoc uses a *single homogeneous* family of numeric index tokens $\langle 0 \rangle, \langle 1 \rangle, \ldots, \langle 63 \rangle$. Both the Grid stage ($\langle \mathrm{grid} \rangle\, G_i\, G_j\, \langle /\mathrm{grid} \rangle$) and the Cell stage ($\langle \mathrm{cell} \rangle\, C_u\, C_v\, \langle /\mathrm{cell} \rangle$) share the same token set, and each token consistently denotes an *absolute* index within its own level. In other words, HiLoc expresses localization as a hierarchical decomposition of coordinates, rather than as an anchor plus a sequence of motion vectors.

Vocabulary size vs. effective resolution. For an effective per-axis resolution K, GETok requires a vocabulary of size $n^2 + m^2 + 1$, in which the grid tokens alone grow *quadratically* with resolution—reaching resolution K demands at least K^2 dedicated grid tokens. HiLoc instead needs only $\max(M, N)$ numeric tokens: thanks to hierarchical reuse between Grid and Cell, 64 shared tokens already yield $K = MN = 64 \times 64 = 4096$. At matched effective resolution, GETok therefore requires $O(K^2)$ specialized tokens while HiLoc needs only $O(\sqrt{K})$ shared tokens, a $64\times$ compression in vocabulary budget (see Proposition 3.1 and its vocabulary-budget remark).

E Limitations

While HiLoc demonstrates significant improvements in spatial localization accuracy and reliability, the framework introduces some limitations that warrant discussion.

Learning Format Complexity. The hierarchical Grid→Cell representation introduces additional complexity in the learning format compared to flat coordinate generation. While the two-level decomposition reduces per-step classification complexity from $O(MN)$ to $O(M + N)$, it requires the model to learn a more structured output grammar with nested tags and sequential dependencies.

This increased format complexity may pose challenges for models with limited capacity or when training data is scarce, as the model must simultaneously master both spatial reasoning and the structured output syntax.

This structured output format, while beneficial for spatial reasoning, requires more sophisticated learning compared to the simpler flat coordinate sequences used in baseline methods. The increased format complexity may lead to longer training convergence times and higher sensitivity to hyperparameter choices during the SFT stage.

To address this limitation, we employ several strategies: *Progressive training*: We first train the model on simpler localization tasks before introducing the full hierarchical structure; *GRPO post-training*: The reinforcement learning stage helps refine the structured output generation.

Despite these challenges, our experiments demonstrate that the benefits of hierarchical spatial reasoning outweigh the increased format complexity, particularly for applications requiring high precision and reliability.

F Data Engine for HiRef-500K: Full Details

We provide the complete description of the data engine used to construct HiRef-500K (Section 3.4).

F.1 Group Construction

The generation operates at the *image–category group* level. Given the LVIS annotation file, we group all instances by $(\texttt{image_id}, \texttt{category_id})$ pairs. For each group, we collect the annotation IDs, bounding boxes (in $[x, y, w, h]$ format).

F.2 Structured Prompting

For each group, we construct a structured prompt that provides the teacher model with: (i) the category name; (ii) a numbered list of candidate instances, each annotated with its `ann_id`, bounding box; and (iii) detailed generation instructions requesting diverse expression types. The prompt explicitly requests K expressions per group (default $K{=}10$) and enforces five expression types:

- **Attribute**: descriptions based on visual properties (e.g., "*red ball*", "*blue wooden chair*");
- **State**: descriptions based on object states (e.g., "*dog running on the grass*", "*person wearing a black leather jacket*");
- **Relation**: spatial/relational descriptions (e.g., "*person standing to the right of the car*", "*book placed behind the laptop*");
- **Set logic**: set-based descriptions (e.g., "*all red objects in the scene*", "*cups that are not on the table*");
- **Nonexist**: descriptions with no matching instance (e.g., "*person wearing a red hat*", "*wooden furniture*"), providing training signal for the `<no_box>` rejection mechanism.

The prompt includes style references for each type to guide the teacher toward concrete, discriminative language. For single-target expressions, the prompt requests combining at least two visual properties to ensure discriminative precision. For multi-target expressions, it encourages shared-property or set-based descriptions. The prompt also instructs the teacher to leverage spatial/relational language (left of, right of, above, below, next to, behind, in front of) when possible.

F.3 Generation and Parsing

We use Qwen3-VL-32B-Instruct as the teacher model, with temperature 0.2 and max 512 new tokens per group. The model receives both the image and the structured text prompt, and is instructed to return a JSON array where each item contains:

- `text`: the referring expression in English;
- `expr_type`: one of {attribute, state, relation, set_logic, nonexist};

- `target_ann_ids`: a list of annotation IDs from the provided candidates (empty for nonexist);
- `hardness`: one of {easy, medium, hard}.

Each parsed expression is annotated with its cardinality (*zero*/*one*/*many*) based on the number of target IDs. This cardinality annotation is critical for training:

- **Zero-target** expressions train the `<no_box>` rejection behavior;
- **One-target** expressions train precise single-instance grounding;
- **Many-target** expressions train multi-instance detection.

F.4 Dataset Statistics

Figure 9 presents visual examples illustrating the diversity of referring expressions in the HiRef-500K dataset, while Figure 10 provides a statistical analysis of the dataset generated by our data engine. (a) shows the proportion of instances per phrase in the annotations, i.e., the percentage of phrases that correspond to 0, 1, 2, or more than 2 instances, reflecting the cardinality distribution of the dataset. (b) shows the distribution of annotation text lengths, illustrating the diversity of expression complexity in the generated referring expressions.

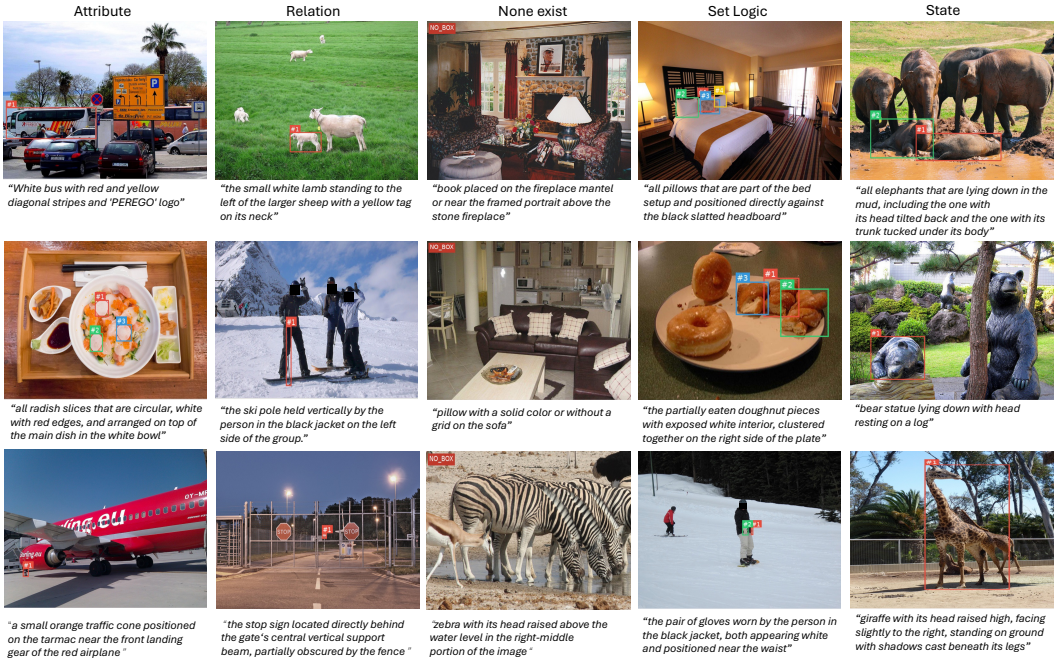
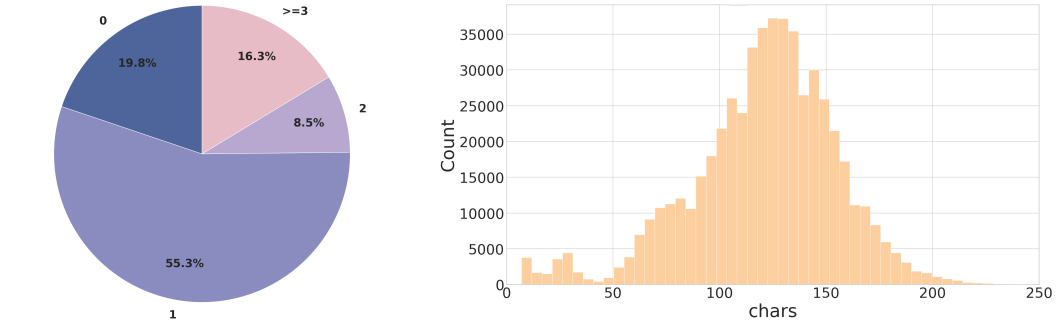


Figure 9: Visual examples from the HiRef-500K dataset, illustrating the diversity of referring expressions across different expression types and object categories. The images are obtained from LVIS[14].

G Additional Qualitative Comparisons

To complement the quantitative results in the main paper, we provide additional visual comparisons between HiLoc and representative baselines on two core tasks: object detection and referring expression grounding. These examples are intended to qualitatively illustrate the behavioral differences induced by the hierarchical spatial indexing and structured output design.

Figure 11 compares detection results across a variety of scenes, including dense and multi-scale settings. HiLoc tends to produce tighter bounding boxes and more complete coverage of small or



(a) Target Annotations IDs Length Share

(b) Text Length Distribution

Figure 10: Statistical analysis of the HiRef-500K dataset. (a) Proportion of instances per phrase: the percentage of phrases corresponding to 0, 1, 2, or >2 instances. (b) Distribution of annotation text lengths across the generated referring expressions.

clustered instances, while reducing duplicate and mis-localized predictions commonly observed in the baselines.

Figure 12 shows comparisons on referring expression tasks involving attributes, relations, and cardinality constraints. HiLoc's outputs align more faithfully with the textual description, and the `<no_box>` behavior is correctly triggered when the referred object is absent, reflecting the benefits of the structured decoding format.

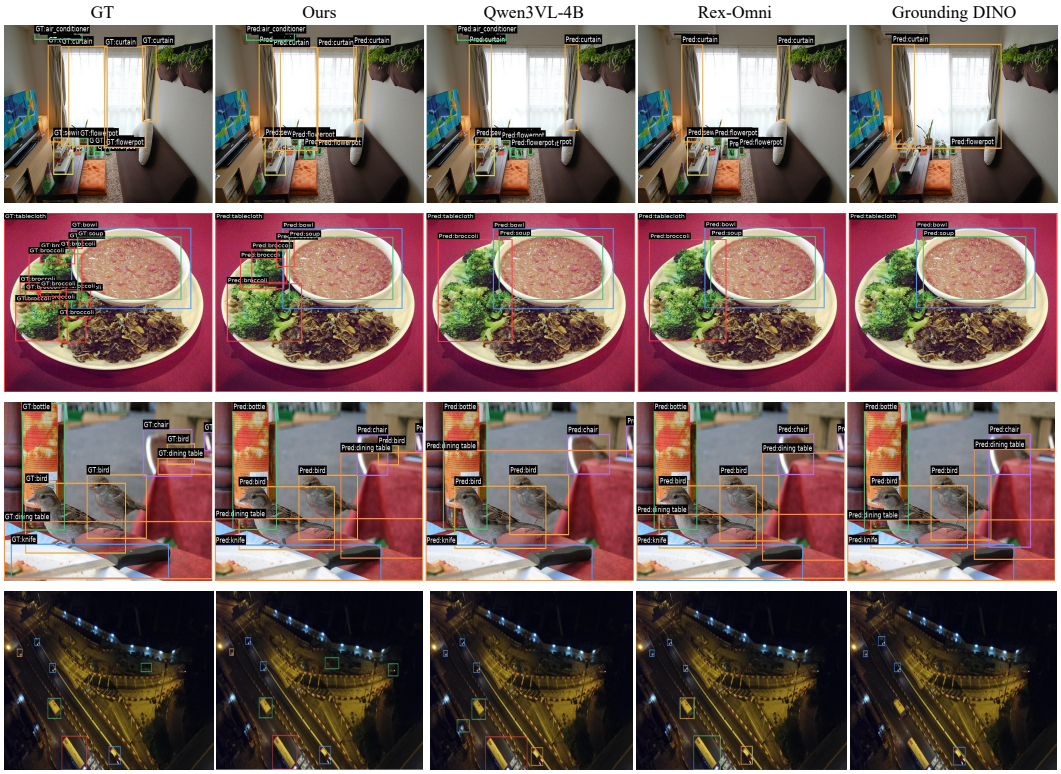


Figure 11: Visual comparison of detection performance: HiLoc vs. baseline methods across diverse object detection scenarios, demonstrating improved precision and recall.

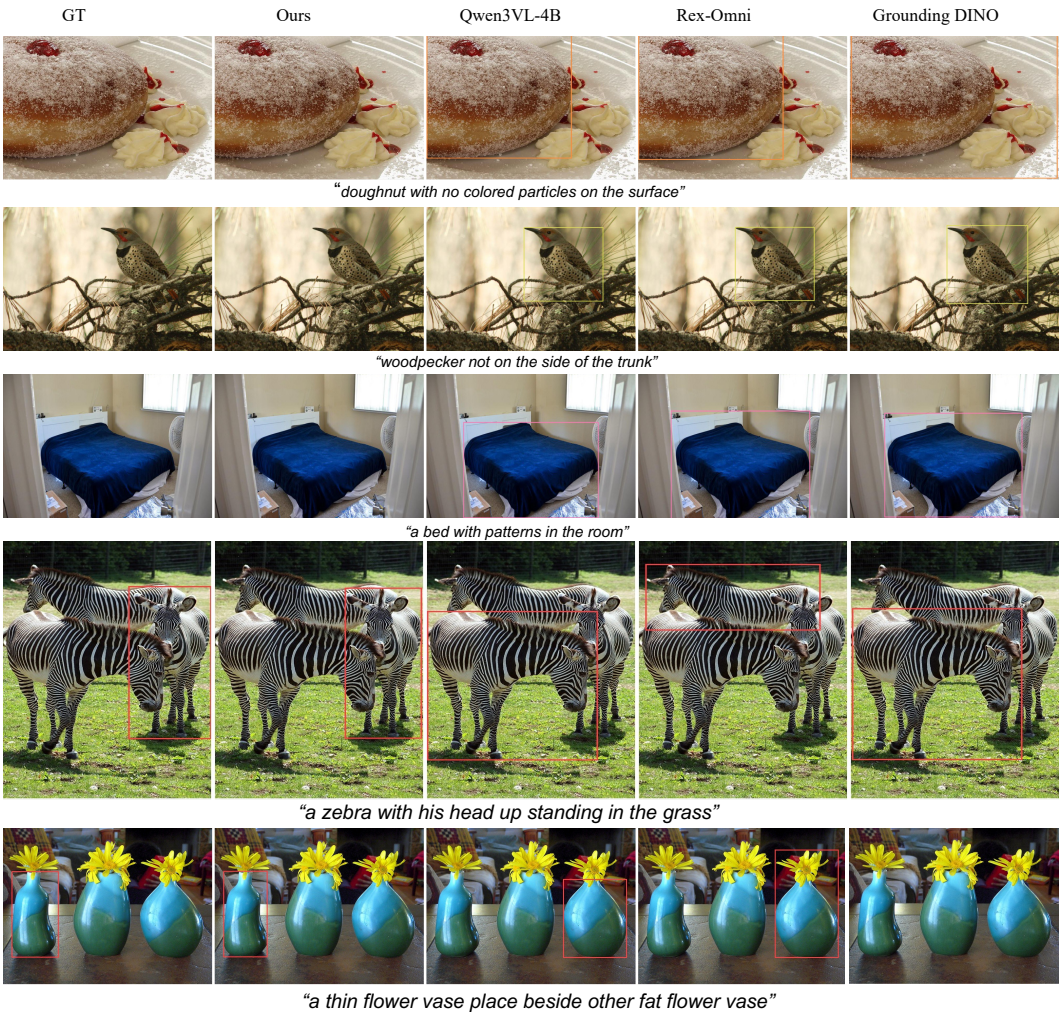


Figure 12: Visual comparison of grounding performance: HiLoc vs. baseline methods in referring expression comprehension tasks, showing more accurate spatial alignment with textual descriptions.

H Formal Analyses and Assumptions

We clarify the scope and assumptions of the original statements in Section 3.2 and Appendix A. The corrected bounds and qualifications below supersede the unconditional interpretations of those statements. They concern coordinate discretization and valid index predictions, not an unconditional guarantee about the learned model.

H.1 Quantization and Decoder Convention

For $x \in [0, W)$, define $q = \lfloor Kx/W \rfloor$, $g = \lfloor q/N \rfloor$, and $c = q \bmod N$, where $K = MN$. Clip the endpoint $x = W$ to the last bin. Both flat and hierarchical encodings identify the same fine bin, with $q = Ng + c$. A decoder can select its lower edge $\hat{x}_{\text{edge}} = qW/K$ or center $\hat{x}_{\text{center}} = (q+1/2)W/K$.

Derivation of Proposition 3.1. The true coordinate is in $[qW/K, (q+1)W/K]$. Its maximum distance from the center is $W/(2K)$; its distance from the lower edge is at most W/K. Under a uniform within-bin residual, the mean absolute errors are $W/(4K)$ and $W/(2K)$, respectively. Thus a lower-edge formula must not be paired with a center-decoding worst-case bound. At fixed K, the representations have identical quantization error under the same decoder. At fixed token budget L, flat indexing provides $K = L$, while shared two-level indexing provides $K = L^2$ when $M = N = L$. Substitution gives the respective fixed-budget bounds $W/(2L)$ and $W/(2L^2)$. □

These are discretization errors with correct indices. They neither bound total model error nor guarantee an accuracy increase when resolution is increased. Structural tags are not counted in L.

H.2 Per-Step Classification versus Total Loss

Derivation of Proposition 3.2. For a uniform C-class distribution, the negative log-probability of the target is $\log C$. Applying this to the valid Grid and Cell classes gives $\log M$ and $\log N$, whose sum equals $\log(MN)$, the flat coordinate loss at the same resolution. For $M = N = L$, each step has L candidate coordinate indices rather than L^2, but requires two predictions per axis rather than one. □

The training objective operates over the extended language vocabulary. The conditional calculation does not assume or establish that implementation masks all non-coordinate logits. Language/structural-token probabilities, sequence length, model features, and parameter sharing all affect the actual loss and optimization.

For a conditional softmax with C classes, $\partial \ell / \partial z_j = p_j - \mathbf{1}[j = y]$. At uniform initialization, the incorrect-class squared gradient sum is $(C-1)/C^2$. Those gradients are useful learning signals, not independent noise, so a ratio formed from them does not establish training stability. We make no $O(C \log C)$ convergence-rate claim. Figure 7(c) tracks newly initialized coordinate-token embeddings; its observed curves are an empirical diagnostic, not a proof of convergence or conditioning.

H.3 Conditional Error Locality

Derivation of Proposition 3.3. Partition predictions into disjoint events: a wrong Grid (probability p_g), a correct Grid and wrong Cell (probability $(1-p_g)p_c$), and two correct indices (probability $(1-p_g)(1-p_c)$). With valid indices and a decoder inside the predicted bin, any coordinate error is bounded by W. Conditioned on a correct Grid, prediction and target belong to the same interval of width W/M, regardless of the Cell-index offset. If both indices are correct, only the decoder-specific quantization bound δ_q remains. The law of total expectation gives the corrected bound $\mathbb{E}|\hat{P}_w - P_w| \le p_g W + (1-p_g)p_c W/M + (1-p_g)(1-p_c)\delta_q$, rather than the unconditional bound in Eq. (3). Defining p_c conditionally avoids an independence assumption. □

More precisely, with $\Delta g = \hat{g} - g$ and $\Delta c = \hat{c} - c$,

$$|\hat{x} - x| \le \frac{W}{M}|\Delta g| + \frac{W}{MN}|\Delta c| + \delta_q. \tag{15}$$

This triangle bound makes the role of actual index distances explicit. A Cell mistake may span multiple fine cells, and a Grid mistake may span multiple Grids. Cross-entropy does not imply

exponentially rare distant Grid errors. Likewise, a decimal leading-digit change can produce different displacements depending on its magnitude; it is not a universal $0.1W$ error. We do not claim an unconditional expected-error advantage over all flat decoders.

I Supplementary Controlled Experiments

The resolution and matched-training controls are reported in Section 4.4. Here we provide the complementary token-sharing study.

I.1 Hierarchical Grouping and Token Sharing

We extract single-category, single-instance samples from the COCO training and validation sets for a controlled grounding study. Training data and configuration are identical across variants. Shared-64 is HiLoc with one set of 64 index tokens used by both Grid and Cell. Separate-64 uses distinct sets <g0>–<g63> and <c0>–<c63>, totaling 128 coordinate tokens. Both use $M = N = 64$.

Base-64 factorizes each discrete coordinate as $q_x = 64x_{\text{high}} + x_{\text{low}}$ and $q_y = 64y_{\text{high}} + y_{\text{low}}$, but serializes digits in axis order without Grid/Cell grouping. The symbolic point formats are:

```
Shared-64:
<point><grid><x_high><y_high></grid>
       <cell><x_low><y_low></cell></point>
Base-64:
<point><x_high><x_low><y_high><y_low></point>
```

Here the high/low placeholders denote actual numeric index tokens. Both Shared-64 and Base-64 have $L = 64$ and $K = 4096$. Base-64 changes the serialization order and removes the Grid/Cell tags, so the experiment tests this combined grouping/serialization choice rather than isolating an abstract hierarchy independently of syntax.

Table 8: COCO single-category, single-instance grounding study. All variants have $K = 4096$. F1 is reported in percent; these subset results are not directly comparable to the full-benchmark tables.

Representation	L	F1 at IoU 0.5	F1 at IoU 0.95	Mean F1
Shared-64	64	79.2	4.9	45.4
Separate-64	128	78.0	2.9	38.0
Base-64	64	46.2	1.4	15.4

Shared-64 improves mean F1 by 7.4 points over Separate-64 and by 30.0 points over Base-64 in this setting. These results support the chosen shared-token and Grid/Cell serialization design on this subset; they do not establish the same effect size for all tasks or constitute a GETok comparison.

J Additional Related-Work Context and Limitations

Griffon [54] constructs non-existing queries using LVIS negative-category annotations and uses GPT-4V to diversify negative referents through attributes such as color and position. Thus, LVIS-based rejection data and model-generated descriptions are established ideas. HiRef-500K organizes annotation-linked expressions into five types, including positive-target set logic and zero-target non-existence. Its role is to support training rather than establish a new negative-sample construction principle. GETok [35] also uses coarse spatial tokens and refinement; Appendix D discusses the representational differences without claiming a matched empirical comparison.

Scope of the comparison. GETok combines n^2 grid anchors with local offset operations [35]. Its refined precision depends on the anchor spacing, offset step, and refinement sequence, so it cannot generally be equated with an unrefined $K \times K$ anchor vocabulary. We therefore make no $O(K^2)$ versus $O(\sqrt{K})$ matched-precision claim against GETok. The Shared/Separate ablation in Appendix I.1 tests HiLoc's own token sharing; it is not a GETok reproduction. We report no head-to-head GETok result.

Attribution and generalization. The main benchmark tables compare complete systems with different backbones, training mixtures, prompts, and optimization recipes. Controlled same-backbone experiments provide the representation-specific evidence. Equal-resolution results isolate a resolution confound, but the matched Flat-1000 comparison still differs in effective resolution. Cross-backbone transfer of the measured gains remains unestablished here. HiLoc also trails Rex-Omni at VisDrone F1 at IoU 0.5 (59.9 versus 61.6), despite higher scores at stricter overlap thresholds.

Inference cost and prompting. HiLoc adds hierarchical indices and structural tags to the output sequence. We do not report latency, throughput, or scaling with instance count, so fewer coordinate-token types should not be read as a measured decoding-speed improvement. The fixed training template does not by itself establish robustness to unrestricted instructions.

Data and statistical scope. Annotation-linked targets and construction checks do not eliminate all ambiguous or hallucinated descriptions. A quantitative semantic-quality audit remains a limitation. The reported experiments are not accompanied by multi-run uncertainty estimates; small score differences should not be interpreted as statistically established improvements.

Theory. Quantization bounds depend on the decoding convention and comparison budget. Coordinate-class counts do not guarantee easier optimization. Cell errors are local only conditional on a correct Grid, and Grid errors need not be rare or nearby.

K Clarifications of HiRef-500K Construction

The following construction details clarify the original description in Appendix F. In particular, the final generation request is five expressions per group, and only non-existence expressions have zero targets.

K.1 Group Construction

The generation operates at the *image–category group* level. Using LVIS v1 train (`lvis_v1_train.json`) and available COCO train2017 images, we group instances by $(\texttt{image_id}, \texttt{category_id})$ pairs. We retain groups with at least three same-category instances, remove missing or unreadable images, and sort annotations by ID. At most 24 candidate instances, their IDs, bounding boxes (in $[x, y, w, h]$ format), and relative positions are supplied to the generator. LVIS validation data are not used for training.

K.2 Structured Prompting

For each group, we construct a structured prompt that provides the teacher model with: (i) the category name; (ii) a numbered list of candidate instances, each annotated with its `ann_id`, bounding box; and (iii) detailed generation instructions requesting diverse expression types. The prompt requests five expressions per group and enforces five expression types:

- **Attribute**: descriptions based on visual properties (e.g., "*red ball*", "*blue wooden chair*");
- **State**: descriptions based on object states (e.g., "*dog running on the grass*", "*person wearing a black leather jacket*");
- **Relation**: spatial/relational descriptions (e.g., "*person standing to the right of the car*", "*book placed behind the laptop*");
- **Set logic**: set-based descriptions (e.g., "*all red objects in the scene*", "*cups that are not on the table*");
- **Nonexist**: descriptions with no matching instance (e.g., "*person wearing a red hat*", "*wooden furniture*"), providing training signal for the `<no_box>` rejection mechanism.

The prompt includes style references for each type to guide the teacher toward concrete, discriminative language. For single-target expressions, the prompt requests combining at least two visual properties to ensure discriminative precision. For multi-target expressions, it encourages shared-property or set-based descriptions. The prompt also instructs the teacher to leverage spatial/relational language (left of, right of, above, below, next to, behind, in front of) when possible.

K.3 Generation and Parsing

We use Qwen3-VL-32B-Instruct as the teacher model, with temperature 0.2 and max 512 new tokens per group. The model receives both the image and the structured text prompt, and is instructed to return a JSON array where each item contains:

- `text`: the referring expression in English;
- `expr_type`: one of {attribute, state, relation, set_logic, nonexist};
- `target_ann_ids`: a list of annotation IDs from the provided candidates (empty for nonexist);
- `hardness`: one of {easy, medium, hard}.

Only non-existence expressions have zero targets. Attribute, state, relation, and set-logic expressions have one or more targets. In particular, "person without a helmet" denotes visible people satisfying a negative constraint, not necessarily an absent target. Each parsed expression is annotated with its cardinality (*zero*/*one*/*many*) based on the number of target IDs. This cardinality annotation is critical for training:

- **Zero-target** expressions train the `<no_box>` rejection behavior;
- **One-target** expressions train precise single-instance grounding;
- **Many-target** expressions train multi-instance detection.

K.4 Parsing and Target Validation

Generated JSON is normalized, target IDs must form a list, and cardinality is recomputed from annotation-linked targets. Failed, empty, or unparsable generations are regenerated with single-target, multi-target, or non-existence templates. Target boxes are retrieved from the original annotations rather than generated as free-form coordinates. Malformed structures and invalid target mappings are checked during conversion. The intended final annotation contract is that only non-existence examples have an empty target set; a failed mapping is not itself evidence of visual absence.